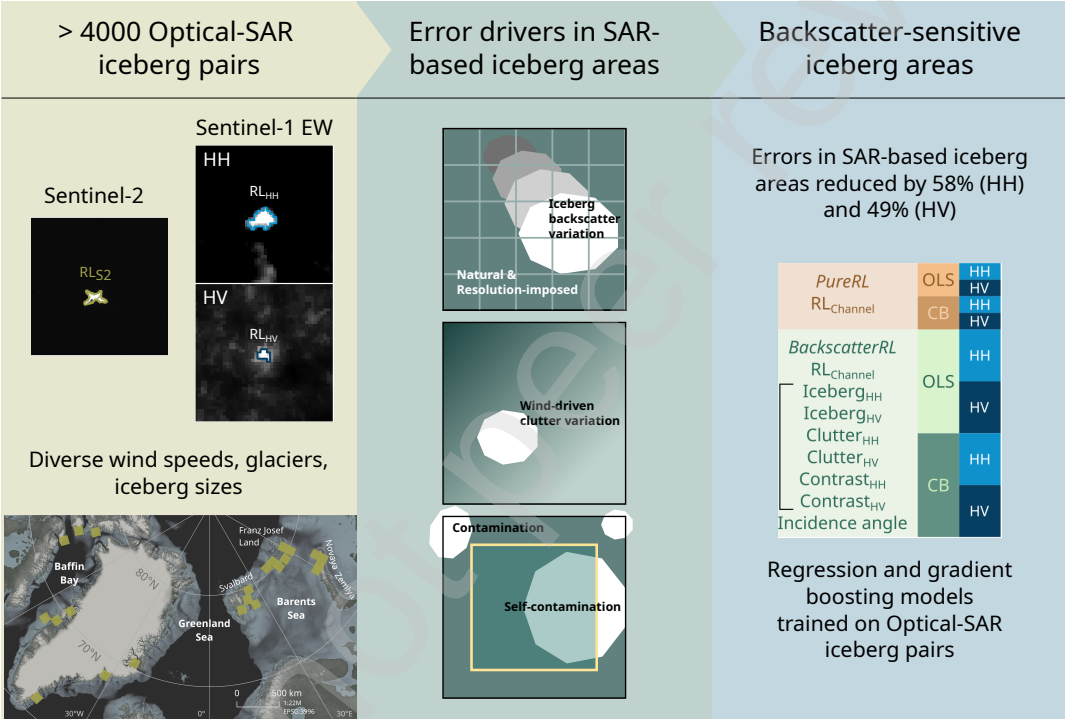


Graphical Abstract

Backscatter-sensitive retrieval of iceberg areas from Sentinel-1 Extra Wide Swath SAR data

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Highlights

Backscatter-sensitive retrieval of iceberg areas from Sentinel-1 Extra Wide Swath SAR data

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- Dataset of more than 4000 icebergs matched between co-located optical and SAR data across the Arctic in summer months from 2016 to 2023.
- New method predicts more accurate iceberg areas from Sentinel-1 Extra Wide Swath data by utilizing ocean and iceberg backscatter statistics, and the incidence angle.
- A constant false alarm rate algorithm primarily overestimates iceberg areas, with error variations driven by a variable ocean and iceberg backscatter.

Backscatter-sensitive retrieval of iceberg areas from Sentinel-1 Extra Wide Swath SAR data

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Abstract

We present the first systematic study on Arctic iceberg area retrieval from Sentinel-1 Extra Wide Swath SAR data. Our dataset contains 4014 Arctic icebergs in open water. Icebergs were detected and matched in co-located Sentinel-1, and optical Sentinel-2 acquisitions from May to September in the years 2016 to 2023. To retrieve iceberg areas from Sentinel-1 data, we initially apply a **constant false alarm rate (CFAR) algorithm**. The Sentinel-1 iceberg areas are moderately correlated with the Sentinel-2 reference iceberg areas, with Pearson's r-values of 0.65 at HH- and 0.69 at HV-polarization. The CFAR algorithm mostly overestimates iceberg areas in both channels. The 10th and 90th percentiles of the relative error are -36% and +569% at HH-, and -37% and +349% at HV-polarization. In addition to a resolution-imposed size-dependency of the error, we find that the HH ocean clutter is moderately negatively correlated with the relative error (Pearson's r-value: -0.64). The CFAR algorithm may underestimate iceberg areas in the HV channel when the iceberg backscatter is low, but the algorithm overestimates

areas at any iceberg backscatter. Additionally, we find weak correlations between the HH (HV) iceberg backscatter and the Sentinel-2 iceberg area, with a Pearson's r-value of 0.34 (0.26). We utilize these relationships to predict iceberg areas from Sentinel-1 data, using a gradient boosting regression. **Backscatter-sensitive models yield more accurate iceberg areas than backscatter-insensitive models.** In the HH (HV) channel, these novel models reduce the mean absolute error by 58% (49%). Backscatter-sensitive iceberg areas will foster consistent SAR-based observations of Arctic iceberg areas during summer months. Future studies need to **quantify the combined impact of iceberg area retrieval and iceberg detection performance on area distributions and total iceberg areas.**

Keywords: Iceberg areas, Sentinel-1, Sentinel-2, Constant false alarm rate, Arctic

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1. Introduction

Observed size-frequency distributions of icebergs are important to estimate freshwater fluxes (Duprat et al., 2016; Moon et al., 2018; Moyer et al., 2019; Rezvanbehbahani et al., 2020; Shiggins et al., 2023), to advance the understanding of variations in iceberg calving (Bigg et al., 1997; Enderlin et al., 2014; Amaral et al., 2020; Alley et al., 2023), and to initiate iceberg drift and decay models (Keghouche et al., 2010; Marchenko et al., 2019; Monteban et al., 2020).

Arctic icebergs originate from tidewater glaciers in six source regions

across southeast Alaska, the Canadian Arctic archipelago, Greenland, Svalbard, Franz Josef Land and Novaya Zemlya (Bigg, 2015). The Greenland ice sheet generates the largest proportion of the Arctic iceberg population of between 96 and 659 km³ · yr⁻¹ (Bigg et al., 2014). Iceberg areas typically follow power-law distributions near the glacier terminus, and log-normal distributions further away from the terminus (Dowdeswell et al., 1992; Bigg et al., 1996; Kechouche et al., 2010; Kirkham et al., 2017; Sulak et al., 2017; Scheick et al., 2019; Rezvanbehbahani et al., 2020). Satellite data are crucial for quantifying above-waterline iceberg areas, as they are acquired systematically in remote Arctic regions. Studies focusing on quantifying geophysical parameters, such as the iceberg freshwater volume, often employ optical satellite data (Sulak et al., 2017; Scheick et al., 2019; Rezvanbehbahani et al., 2020). SAR satellite data are independent of weather and sun, but the technical hurdle to using SAR data is high. Moreover, the iceberg area retrieval appears to inherit large errors from ambiguous iceberg and ocean backscatter signals (Williams et al., 1999; Buus-Hinkler et al., 2016; Bailey et al., 2024), complicating the take-up of SAR for retrieving iceberg areas. In this paper, we connect iceberg area retrievals from SAR data to retrievals from optical data in a novel way. The study features a large dataset of above-waterline iceberg areas in open water across the Arctic, retrieved from near-coincident Sentinel-1 Extra Wide Swath (EW) SAR data, and optical Sentinel-2 data.

1.1. SAR scattering of icebergs

The manifold shapes, surface roughness and properties of icebergs result in a variable SAR backscatter. Fig. 1 depicts scattering mechanisms of an iceberg in open water. Icebergs primarily exhibit surface and volume scat-

tering (Lewis et al., 1987; Haykin et al., 1994; Power et al., 2001). Double
bounces occur particularly at the edge with adjacent specular reflection on
the water surface (Akbari and Brekke, 2018; Bailey et al., 2024). C-band
microwave radiation can penetrate icebergs up to 13 meters deep (Haykin
et al., 1994; Power et al., 2001; Langley et al., 2007). The contribution of
volume scattering increases with the amount of air inclusions and the crystal
size (Wesche and Dierking, 2012). A strong iceberg backscatter composed
of surface and volume contributions is expected under freezing conditions,
whereas liquid water on the surface dampens the backscatter (Young et al.,
1998; Williams et al., 1999; Power et al., 2001; Wesche and Dierking, 2012;
Dierking and Wesche, 2014; Marino et al., 2016; Færch et al., 2022, 2024). A
higher surface roughness and more complex shape enhance the surface scat-
tering (Young et al., 1998; Power et al., 2001; Wesche and Dierking, 2012).
The iceberg backscatter usually exceeds the backscatter of surrounding open
water at low wind speeds when water primarily exhibits specular reflection.
The surface scattering of open water increases when wind roughens the wa-
ter, particularly at low incidence angles (Young et al., 1998; Power et al.,
2001; Wesche and Dierking, 2012; Barbat et al., 2019).

1.2. Iceberg detection and area retrieval from SAR data

Iceberg detection refers to automatically identifying icebergs in satellite
data. Iceberg area retrieval describes the retrieval of above-waterline ice-
berg areas. In SAR data, we face complex situations of varying wind speeds,
iceberg properties, the spatial resolution, polarization channels, noise level,
speckle, and the incidence angle (Wesche and Dierking, 2012; Buus-Hinkler
et al., 2016; Tao et al., 2016; Akbari and Brekke, 2018; Soldal et al., 2019;

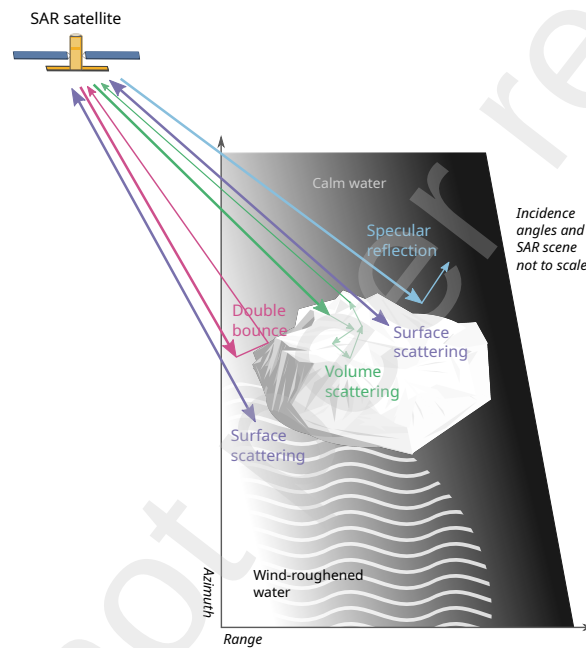


Figure 1: SAR scattering mechanisms of an iceberg in wind-roughened water, and in calm water. Icebergs can exhibit volume scattering, surface scattering, and double bounces. Wind-roughened water shows surface scattering, whereas specular reflection is prominent in calm water.

60 Færch et al., 2023; Bailey et al., 2024). Fig. 2 shows icebergs captured by
 61 Sentinel-1B in complex ocean clutter. Constant false alarm rate (CFAR)
 62 algorithms tackle ambiguous SAR signatures by deriving local thresholds to
 63 distinguish icebergs from complex surroundings (Power et al., 2001; Buus-
 64 Hinkler et al., 2014; Frost et al., 2016; Færch et al., 2023; Bailey et al., 2024).
 65 The operational iceberg mapping in Greenland relies on a CFAR algorithm
 66 (Buus-Hinkler et al., 2014). CFAR uses a moving window where the central
 67 backscatter measurement is tested against the surrounding background. In
 68 open water, the background consists of *ocean clutter* driven by a varying sur-
 69 face roughness that depends on wind speed. The CFAR algorithm separates
 70 an outlier based on the probability of a measurement to belong to the clutter
 71 backscatter distribution (Power et al., 2001; Soldal et al., 2019; Færch et al.,
 72 2023; Bailey et al., 2024). In related approaches, segmentation algorithms
 73 have been developed to cluster ambiguous SAR signatures into internally
 74 homogeneous regions (Silva and Bigg, 2005; Mazur et al., 2017; Akbari and
 75 Brekke, 2018; Barbat et al., 2019; Koo et al., 2023; Shankar et al., 2023).
 76 Tao et al. (2016) combined a segmentation approach with a CFAR algorithm
 77 for ship detection. Convolutional neural networks have been employed to
 78 distinguish Arctic icebergs from ships (Li et al., 2019; Song et al., 2021;
 79 Heiselberg et al., 2022). These different algorithms focus on automatically
 80 detecting the positions of icebergs in SAR data, whereas the iceberg areas
 81 retrieved from SAR have received sparse attention in the Arctic. Williams
 82 et al. (1999) found that iceberg areas obtained from C-band SAR data of
 83 the European Remote-Sensing Satellite 1 were overestimated by 20%, with
 84 considerable variations. Buus-Hinkler et al. (2016) observed overestimated

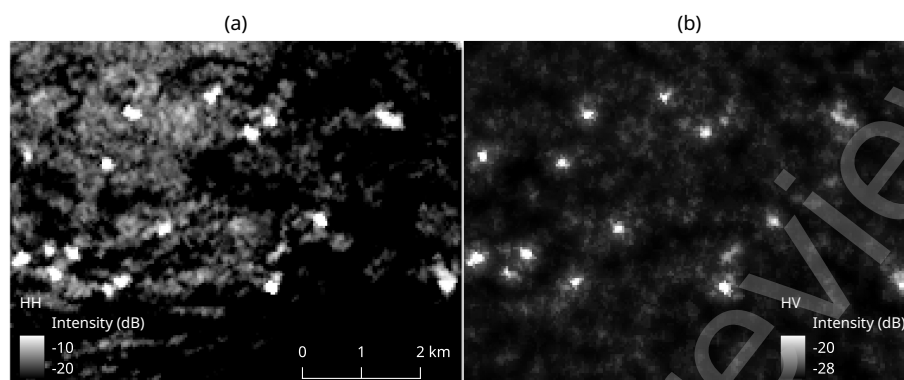


Figure 2: A group of icebergs in the Davis Strait at 76.61957°N, 75.97472°W, captured on 18 June 2021 by Sentinel-1B in the HH channel (a) and the HV channel (b). The wind speed was about 6 m s⁻¹. We observed a heterogeneous sea state in this acquisition.

iceberg areas derived from Sentinel-1 EW data using CFAR algorithms with various parameterizations, compared to optical Landsat data. With a geographically and temporally limited dataset, Bailey et al. (2024) showed weak correlations between the C-band backscatter (RADARSAT-2) and iceberg areas obtained from ground radar. The limited understanding of SAR-based iceberg areas, despite their scientific importance in various applications, motivated our study.

1.3. Scope of the paper

We aimed to improve the Arctic above-waterline iceberg area retrieval from Sentinel-1 EW SAR data, based on a common CFAR iceberg detector. Our analyses are based on several hundred co-located optical Sentinel-2 and Sentinel-1 SAR data that were collected across the entire Arctic from 2016 to 2023. We quantified errors in the Sentinel-1 iceberg areas relative to the Sentinel-2 iceberg areas based on iceberg pairs detected and matched in

99 both datasets. Sentinel-2 data were used as a reference due to their high
100 resolution, and since the iceberg area retrieval from these data has been
101 previously validated (Moyer et al., 2019; Fisser et al., 2024). To understand
102 variables driving the errors in SAR-based iceberg areas, we analyzed the
103 relationship between iceberg and ocean backscatter statistics and the error.
104 Finally, we sought to improve the SAR-based iceberg area retrieval via novel
105 backscatter-sensitive gradient boosting regression models for iceberg areas.

106 2. Data

107 The presented study centers around a dataset of several hundred opti-
108 cal Sentinel-2 acquisitions, and co-located Sentinel-1 EW SAR acquisitions
109 collected across iceberg-populated Arctic waters (Fig. 3). Compiling the
110 dataset required several filtering steps that are described subsequently.

111 2.1. Study area and data selection

112 23 acquisition tiles were selected from the Sentinel-2 acquisition plan
113 across the Barents Sea, the Baffin Bay, and along the western and eastern
114 Greenlandic coast (Fig. 3). As the abundance of icebergs in the Barents Sea
115 is comparably low, more tiles were placed there to obtain a sufficient number
116 of samples. For the investigated years, we focused on the period from May
117 through September, targeting open water and sufficiently high sun altitudes.
118 The Sentinel-1 data were searched co-located with the Sentinel-2 tiles, and
119 acquired at a maximum absolute time lag of eight hours.

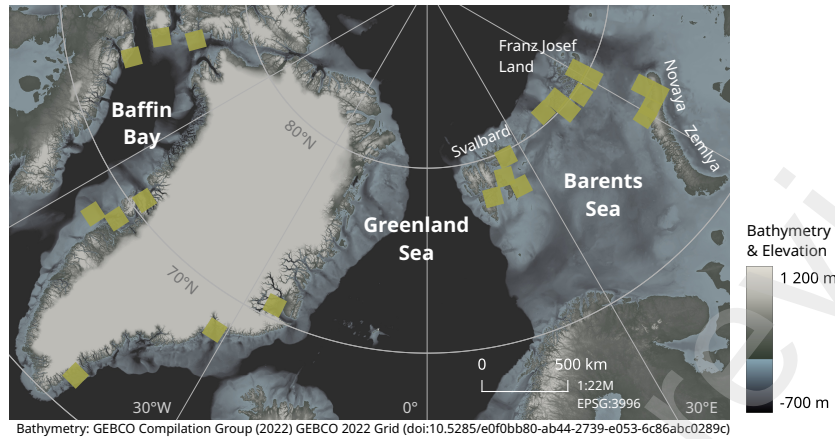


Figure 3: The 23 study areas, delimited by Sentinel-2 acquisition tiles, are distributed across coastal waters in the Baffin Bay, along the west and east coast of Greenland, and among the archipelagoes of the Barents Sea.

2.2. Remote Sensing data

We collected 274 Sentinel-2 acquisitions and 465 Sentinel-1 acquisitions. The number of Sentinel-1 acquisitions exceeds the number of Sentinel-2 acquisitions because often two acquisitions overlapped with the Sentinel-2 tile.

Land pixels were excluded from the analysis using Open Street Map land polygons (OpenStreetMap contributors, 2023) expanded seaward by 2 km to account for inaccuracies in the land polygons, and differences in the image geo-registration.

2.2.1. Sentinel-2

The Copernicus Sentinel-2 mission provides multispectral measurements of reflected solar spectral radiances from two satellites (Tab. 1). We chose Sentinel-2 near-infrared data at 10 m spatial resolution for retrieving reference iceberg areas because the resolution (Tab. 1) is nine times higher than

Table 1: Sentinel-2 data properties.

Swath	Resolution	Bands	Wavelengths
290 km	10 to 60 m	13	443 to 2222 nm

the Sentinel-1 EW GRDM SAR resolution (90 m). One SAR resolution cell thus contains 81 Sentinel-2 pixels. The errors in threshold-based areas retrieved from the near-infrared imagery usually stay between +20% and -20% with the largest errors exhibited by small icebergs (Fisser et al., 2024).

We searched Sentinel-2 level 1C acquisitions with a cloud cover below 2%, and with a solar zenith angle below 65° (Fisser et al., 2024). The digital numbers DN were converted to top-of-atmosphere (TOA) reflectance values ρ by applying the scaling coefficient and an *offset*: $10^{-4} \cdot (DN + \text{offset})$. The *offset* is -1000 for processing baselines ≥ 4.0 , whereas it is 0 prior to that. We follow the Sentinel-2 processing of a previous study without atmospheric correction (Fisser et al., 2024).

2.2.2. Sentinel-1

The Copernicus Sentinel-1 mission is a SAR mission of two satellites, with Sentinel-1B decommissioned since December 2022. The Sentinel-1 satellites carry right-looking, dual-polarized C-band SAR sensors with a center frequency of 5.405 GHz (ESA, 2022b). The Extra Wide Swath mode is in many areas the only, but in all areas the most frequently acquired mode over Arctic coasts. Like Færch et al. (2023), we used the medium resolution ground-range detected (GRDM) EW product (Tab. 2). We chose EW data, although they offer a lower spatial resolution than the Interferometric Wide Swath mode. This has two reasons: First, Buus-Hinkler et al. (2014) ob-

Table 2: Sentinel-1 GRDM EW data properties.

Swath	Incidence angles	Pixel spacing	Resolution
410 km	19 to 47°	40 m	90 m

served more false positive iceberg detections when applying a CFAR iceberg detector to IW data. Moreover, **double bounces** appeared to add inconsistencies to the iceberg area retrieval. Second, EW data are more frequently acquired, and cover larger areas, which is valuable for dense time series of iceberg areas. Furthermore, the coverage increases the number of matches with Sentinel-2 data, advantageous for this study and other studies that leverage Sentinel-1 data in conjunction with optical data.

The EW GRDM data are detected, multi-looked, and projected on an Earth ellipsoid model (ESA, 2022a). For our analysis, we used HH and HV dual-polarized GRDM products.

2.3. Meteorological reanalysis data

We downloaded the Arctic Regional Reanalysis (CARRA) product (Schyberg et al., 2020) from the Copernicus Climate Change Service (C3S) Climate Data Store (2021), providing a 2.5 km in three-hourly timestamps. CARRA relies on the HARMONIE-AROME cycle 40h1.1 with vertical forcing by ERA5 (Køltzow et al., 2022). **The 10 m wind speed and the 2 m air temperature variables** were collected at the timestamps that minimized the time difference to the respective Sentinel-1 acquisition.

172 3. Methods

173 Subsequently, we explain how we compiled our dataset of 4014 Optical-
174 SAR iceberg pairs. Further, we describe the regression models that were
175 designed to predict iceberg areas from Sentinel-1 data.

176 3.1. Iceberg detection and area retrieval - Sentinel-2

177 We filtered remaining clouds in the Sentinel-2 data using a pixel-wise
178 cloud probability calculated by the *s2cloudless* algorithm (Skakun et al.,
179 2022). The cloud probability layer was averaged over a pixel and its four
180 closest neighbors. Pixels were flagged as clouds when the cloud probability
181 exceeded 5%. The percentage of valid observations in the data was recom-
182 puted, and an image was dropped if this value was below 70% (Fig. 4). The
183 near-infrared (NIR) band (B8), here denoted as ρ , was used for the iceberg
184 area retrieval from Sentinel-2 data as previously proposed (Moyer et al., 2019;
185 Fisser et al., 2024).

186 To retrieve iceberg areas from Sentinel-2 data, we used a NIR reflectance
187 threshold of 0.12 that was previously found most accurate for the iceberg
188 area retrieval (Fisser et al., 2024). The Sentinel-2 NIR data are scaled from
189 0 to 1 (0-100% reflectance). Values passing the threshold were flagged as *ice*:

$$f(\rho) = \begin{cases} \text{ice} & \text{if } \rho \geq 0.12 \\ \text{water} & \text{if } \rho < 0.12 \end{cases} \quad (1)$$

190 The *ice* initially encompasses both icebergs and sea ice, where present.
191 This was acceptable here, since we matched icebergs manually. During the

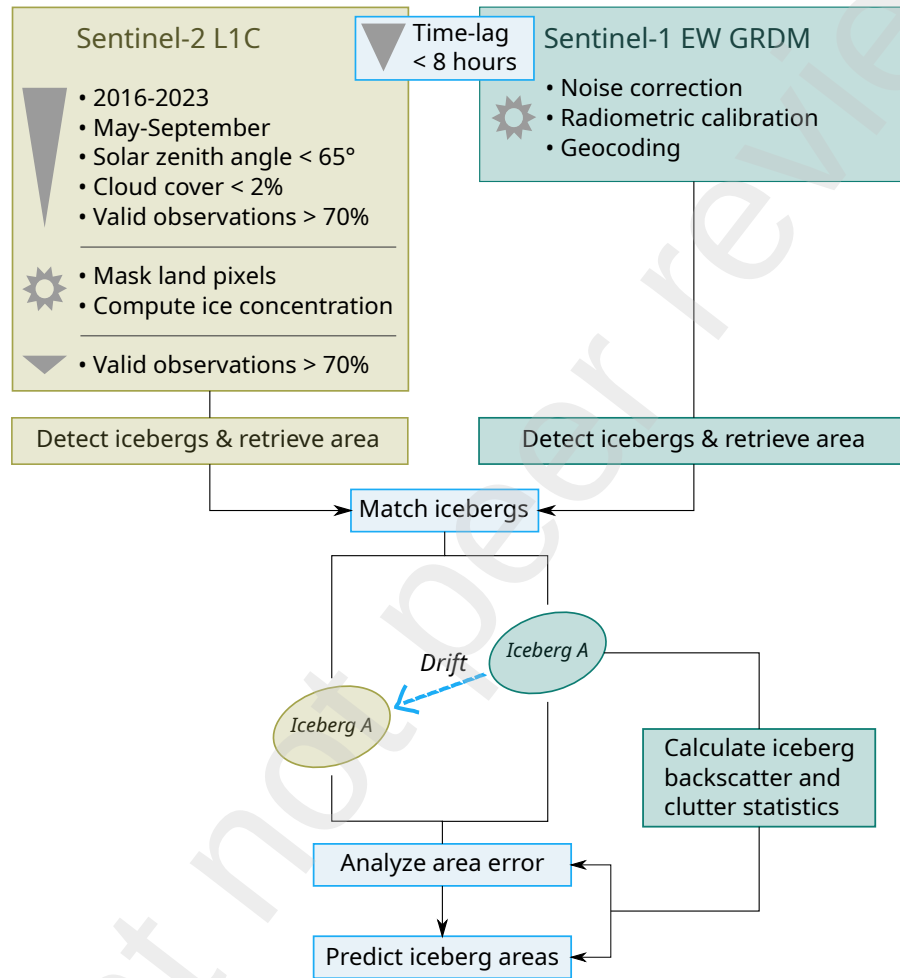


Figure 4: The iceberg dataset was compiled upon near-coincident optical Sentinel-2 acquisitions, and Sentinel-1 SAR acquisitions. Icebergs were matched manually in the acquisition pairs, considering their drift. The error in the Sentinel-1 iceberg areas was analyzed concerning iceberg backscatter and clutter statistics.

192 manual matching, visual inspection of the ice features detected in the optical
193 data enabled us to omit sea ice floes.

194 3.2. Ice filtering - Sentinel-2

195 High ice concentrations, such as mélange, largely prohibit matching indi-
196 vidual icebergs between optical and SAR. We, therefore, excluded high ice
197 concentrations to avoid extensive data processing, given the large size of the
198 remote sensing dataset. Because of the complex coastlines, we opted against
199 passive microwave data for estimating the sea ice concentration. Instead, we
200 calculated an ice concentration (IC) using the Sentinel-2 data. To separate
201 iceberg and sea ice pixels from open water, we used the threshold described
202 above. Initially, the NIR data ρ were divided into 10 km wide squared blocks.
203 The percentage of ρ pixels surpassing 0.12 (reflectance) was calculated in each
204 block with N pixels:

$$IC = 100 \cdot \frac{\sum_{i=0}^N \text{bool}(\rho_i \geq 0.12)}{N}. \quad (2)$$

205 We excluded blocks from the analysis when IC was $\geq 15\%$. The usable
206 fraction of the Sentinel-2 tile was then recalculated. When less than 30% of
207 the Sentinel-2 tile were available for analysis, the Optical-SAR image pair
208 was dropped (Fig. 4).

209 3.3. Iceberg detection and area retrieval - Sentinel-1

210 Since noise in the HV channel may lead to false iceberg detections, we ap-
211 plied the thermal and texture noise correction by the Nansen Environmental
212 and Remote Sensing Center (Park et al., 2018, 2019; Korosov et al., 2022)
213 to the HV channel. The data of both polarization channels were calibrated

214 radiometrically to linear intensities sigma nought (σ_0) and geocoded in the
215 polar stereographic coordinate reference system.

216 Constant false alarm rate algorithms detect icebergs by comparing the
217 radar signal of a given pixel in the SAR image to the backscattering of the
218 local surroundings, here denoted as clutter. CFAR uses a moving window to
219 test every pixel against the clutter distribution that is extracted in an outer
220 window (Fig. 5). The clutter statistics exclude the direct surrounding of the
221 pixel, which is masked by the guard window. A probability density function
222 (PDF) is fitted based on the clutter statistics. The pixel is flagged as an
223 outlier when the probability P that a pixel belongs to the clutter PDF falls
224 below a user-specified probability of false alarm (PFA):

$$f(P, PFA) = \begin{cases} \text{Outlier} & \text{if } P \leq PFA \\ \text{No outlier} & \text{if } P > PFA \end{cases} \quad (3)$$

225 We used a gamma PDF in our study, since Færch et al. (2023) found that
226 the gamma PDF was among the best-performing and most stable PDFs for
227 iceberg detection in open water. The Python code provided by Færch et al.
228 (2023) was used here. We applied the CFAR algorithm separately to the lin-
229 ear intensities of the two polarization channels, resulting in one set of icebergs
230 per channel. Færch et al. (2023) found the strongest detection performance
231 for the gamma PDF using a PFA of 10^{-12} . The authors combined the po-
232 larization channels by a boolean *AND* operator after individually detecting
233 outliers in the respective channel. Early in our study, we found that the ob-
234 tained iceberg areas differ fundamentally by polarization channel, suggesting
235 keeping the outputs of both channels separate in the analysis. Further, we

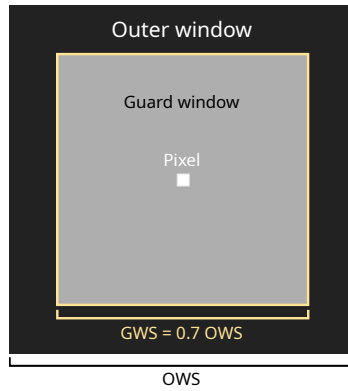


Figure 5: Geometric layout of the CFAR algorithm with an outer window and a guard window centered on a Sentinel-1 pixel at a 40 m pixel spacing.

analyzed the iceberg areas at two different PFA levels of 10^{-6} and 10^{-12} . The high 10^{-6} PFA yields false iceberg detections, but it detects most icebergs present in the data (Færch et al., 2023). Our goal was here to apply the lower PFA of 10^{-6} in the area prediction to decouple the area prediction from the iceberg detection sensitivity.

To investigate how the CFAR outer window size (OWS) and the coupled guard window size (GWS) (Fig. 5) affect the iceberg area retrieval, we tested two outer window sizes, 13 pixels and 49 pixels side length. The GWS is coupled to the OWS with $0.7 \cdot OWS$ (Fig. 5). We chose 49 pixels as the larger OWS because we assumed that the 35-pixel (1400 m) side length of the GWS accommodates most iceberg lengths present in the Arctic.

3.4. Optical-SAR iceberg matching

Most icebergs drifted during the typical time of about five hours between the Sentinel-1 and Sentinel-2 acquisitions. Hence, we searched Optical-SAR iceberg pairs manually by examining several hundred Optical-SAR acqui-

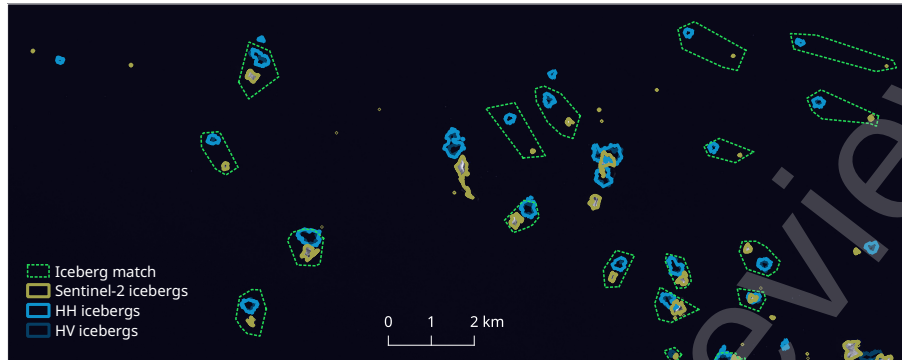


Figure 6: Example of matched icebergs, subject to a 5:15-hour drift. We show a red-green-blue composite overlaid by the detected iceberg outlines, and the iceberg match delineation. The data were acquired on 24 July 2018 over Scoresby Sound in East Greenland at 71.02327°N, 24.52427°W.

tion pairs in a geographical information system. The matching was straightforward when groups of icebergs showed similar drift (example in Fig. 6), but ambiguous drift patterns often impeded any matching. We stored matched icebergs in a database along with their area (optical, SAR), backscatter statistics (iceberg, clutter), incidence angle, and meteorological reanalysis information from the CARRA dataset.

3.5. Iceberg area error

Each Optical-SAR iceberg pair comes with a Sentinel-1 iceberg area A_{Channel} where the channel refers to HH or HV, and a Sentinel-2 iceberg area A_{S2} in square meters. In the analysis, we excluded icebergs with $A_{S2} < 1600 \text{ m}^2$, as these icebergs are smaller than a Sentinel-1 EW GRDM pixel, and we did not expect any sensible area estimate for those icebergs.

To aid the statistical analysis of the iceberg areas, we also used the square root of the areas as the *root length* in meters where applicable, denoted as

exclude
 $A_{S2} < 1600 \text{ m}^2$

265 $RL_{Channel}$, and RL_{S2} . We calculated the relative error (RE) between the
 266 iceberg areas in percentage:

$$RE_A = 100 \cdot \frac{A_{Channel} - A_{S2}}{A_{S2}}. \quad (4)$$

267 Using the RL sometimes simplifies statistical relationships from exponen-
 268 tial to linear. In these cases, we also calculated the RE_{RL} .

269 3.6. Iceberg area models

270 We trained, tested, and validated ordinary least squares regression mod-
 271 els and gradient boosting regression models to predict iceberg areas from
 272 Sentinel-1 data, based on the initial CFAR iceberg delineation and backscat-
 273 ter statistics. The iceberg dataset was divided into a training dataset (75%)
 274 and a validation dataset (25%).

275 3.6.1. Training dataset filtering and balancing

276 Iceberg area distributions are naturally skewed towards small areas, which
 277 is also the case for our dataset. During model testing, we recognized that all
 278 models had difficulties in generalizing across the iceberg area distribution,
 279 with icebergs ranging from about 40 m RL_{S2} to 632 m RL_{S2} . We found
 280 that the skewed area distribution and outliers caused over-fitting to smaller
 281 icebergs, and under-fitting to larger icebergs. The $Bias_{CFAR}$ resembles a
 282 normal distribution with a positive mean and a tail at positive values that
 283 accommodates the outliers. We excluded icebergs with $Bias_{CFAR}$ higher
 284 than two standard deviations above the mean. For the HH (HV) channel, 79
 285 (69) icebergs were dropped from the training dataset.

75% train
25% valid

root length
(RL)
distribution

TEST

TEST
under/over
fit by Area

286 The skewness of the area distribution itself further hindered model gen-
 287 eralization. Imbalanced training data can cause imbalanced predictive skill
 288 in regressions because the low abundance samples have limited influence on
 289 minimizing the error (Torgo et al., 2015; Camacho et al., 2022), i.e., the mod-
 290 els are over-fitted to the abundant samples. To balance training datasets, a
 291 combination of under-sampling high abundance samples and over-sampling
 292 low abundance samples has been proposed (Torgo et al., 2015; Camacho et
 293 al., 2022). Our goal was to compromise between balancing the dataset and
 294 keeping sufficient samples for the model training. To achieve this, Torgo et al.
 295 (2015) and Camacho et al. (2022) created synthetic samples to over-sample
 296 the low abundance samples. We decided to not create any synthetic samples.

297 We divided the dataset into five 22100 m² wide area bins, starting at 1600
 298 m². For the respective polarization channel, the minimum number of icebergs
 299 n among the five bins was calculated. In both channels, n was found in the
 300 area bin > 90000 m². For HH (HV), n was 69 (75). We represented every
 301 area bin by at least $n_{min} = 3 \cdot n$ icebergs, and by a maximum of $n_{max} = 6 \cdot n$
 302 icebergs. Bins with less than n_{min} samples were over-sampled by replicating
 303 all samples as often as they fit into n_{min} . To fill up, additional samples were
 304 randomly drawn until n_{min} was reached. Thus, the area bin > 90000 m² is
 305 replicated three times. The ratio between the most abundant bin and the
 306 least abundant bin is therefore 2:1 in the training dataset. As there is no
 307 recommended value for this ratio, we chose a value that keeps a reasonable
 308 amount of samples in the high abundance bins without too strongly over-
 309 sampling the low abundance bins. The HH (HV) models were trained using
 310 2335 (2540) icebergs, and validated using 804 (750) icebergs. The validation

Imbalanced training data

* synthetic low abundance data?

Quota sampling to n_{min} ?

keep validation data separate?

data was unaffected, as it was separated prior to balancing the dataset. Using the validation data, we compared predictions based on training with and without considering balancing. The balancing decreased the overall performance in terms of the MAE, however, the models generalized better across the iceberg size range. We thus opted for training the models based on the balanced dataset.

3.6.2. Model training and validation

We trained and validated eight models, four per polarization channel - two feature sets, and two regression algorithms (Fig. 7). The models were trained and validated separately per channel, but the features were the same for both channels. The dependent variable in all models is RL_{S2} (Fig. 7). We chose the RL here to reduce the scale-dependency of the error, and due to simpler linear relationships to backscatter statistics.

model training
separate by
channel

The *PureRL* model is a single-variable regression where $RL_{Channel}$ is the only feature. The *BackscatterRL* feature set contains $RL_{Channel}$, six backscatter statistics, and the incidence angle (Fig. 7). We refer to the *BackscatterRL* models as *backscatter-sensitive*. The Contrast (Fig. 7) was calculated pixel-wise as the difference between the $Iceberg_{pixel}$ intensity (dB) and the mean intensity in the outer window OW_{mean} (dB):

$$Contrast = Iceberg_{pixel} - OW_{mean}. \quad (5)$$

Backscatter statistics were calculated in the two channels. The averages of all contrast values within the iceberg outline are $Contrast_{HH}$ and $Contrast_{HV}$. The averages of all iceberg intensities in the iceberg outline are $Iceberg_{HH}$ and $Iceberg_{HV}$. The OW_{mean} statistics are denoted as $Clutter_{HH}$ and $Clutter_{HV}$.

can they do
this if $\neq$ normal
distr. ?

344 corresponding RL_{S2} in the dependent set y . The well-known OLS regression
345 minimizes the mean squared error MSE between x and the predicted $\hat{y}$:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (6)$$

346 where $\hat{y}$ is calculated by a linear regression. The intercept and slope
347 values minimizing MSE were found through a closed-form solution. For the
348 *PureRL* feature set, the regression equation is $RL_{S2} = b + m \cdot RL_{S1}$. For
349 the *BackscatterRL* feature set, slope values were derived for every feature in
350 a matrix of k features and n observations multiplied by a vector of k slope
351 values with a single intercept, aiming to minimize $\sum_{i=0}^n MSE_i$.

352 The CatBoost algorithm expands the original gradient boosting algorithm
353 developed by Friedman (2001). Gradient boosting algorithms optimize a loss
354 function using an ensemble of weak learners. Here, a weak learner is a decision
355 tree that is structured into nodes, and divided by binary splits based on a
356 subset of the training features. Boosting describes how weak learners are
357 added to the ensemble incrementally. The final prediction is the sum of the
358 predictions by all weak learners. The initial guess $F_0(x)$ comes with the loss
359 function $L(y_i, \rho)$ over n observations (Friedman, 2001):

$$F_0(x) = \arg \min_{\rho} \sum_{i=1}^n L(y_i, \rho). \quad (7)$$

360 Here, ρ controls the individual contributions of weak learners. New weak
361 learners $h(x; a_m)$ with a as a set of parameters are added in M steps. To
362 improve the ensemble at step m , the steepest gradient on the loss curve
363 $-g_m(x_i)$ is searched (Friedman, 2001) with reference to the previous predic-
364 tions $F_{m-1}(x)$:

back
scatter
↓

$$-g_m(x_i) = - \left[\frac{\delta L(y_i, F(x_i))}{\delta F(x_i)} \right]_{F(x)=F_{m-1}(x)}. \quad (8)$$

Based on the steepest gradient, the parameters a_m (Eq. 9) of the weak learner are determined. More generically than ρ_m , the learning rate v controls the contribution of weak learners to the ensemble to mitigate overfitting (Friedman, 2001):

$$F_m(x) = F_{m-1}(x) + v \cdot \rho_m h(x; a_m). \quad (9)$$

Ordered boosting, implemented in CatBoost, describes random permutations of the training dataset to evaluate splits in the weak learners, aiming to reduce overfitting. Prokhorenkova et al. (2017) give further details on the algorithm, and we used the provided CatBoost Python package.

The learning rate v is a key hyperparameter in CB and other gradient boosting algorithms (Eq. 9). It can range from 0 to 1 where low values are favorable to enhance model generalization (Friedman, 2001).

We tuned the hyperparameters of each CB model individually via a grid search 5-fold cross-validation (Tab. 3). Across all iterations, the average of the mean absolute error (MAE) was calculated based on the test portion as MAE_{Test} . Additionally, using the validation data that was kept separate from the training data, $MAE_{Validation}$ was calculated for a given combination of hyperparameters. To avoid overfitting, we chose hyperparameters that minimized the absolute difference between MAE_{Test} and $MAE_{Validation}$.

For evaluating the models, we calculated the *Bias*, the MAE, the relative error, and Pearson's r-values. The *Bias* is defined as:

Table 3: Tested and chosen hyperparameter values for **CB models**.

	Tested values	PureRL		BackscatterRL	
		HH	HV	HH	HV
Iterations	{500, 1000}	500	500	500	500
Learning rate	{0.1, 0.15, 0.25}	0.1	0.1	0.1	0.1
Depth	{4, 8}	4	4	4	4

$$Bias = A_{Predicted} - A_{S2}, \quad (10)$$

and the MAE is the mean over the absolute biases. To assess how the area models impact the $Bias$, we calculated Δ_{Bias} :

$$\Delta_{Bias} = Bias - Bias_{CFAR} \quad (11)$$

where $Bias_{CFAR}$ is the bias in the initial CFAR-based iceberg areas.

4. Results

Our study features a large and diverse dataset of Optical-SAR iceberg pairs. Based on these iceberg pairs, we quantified and analyzed errors in SAR-based iceberg areas with reference to the optical counterparts. Further, we evaluated the iceberg area models that aimed to predict more accurate iceberg areas from Sentinel-1 EW data.

4.1. Iceberg dataset

The dataset contains 4014 iceberg pairs collected in West Greenland (35%), in the Baffin Bay (31%), in East Greenland (28%), and in the Barents

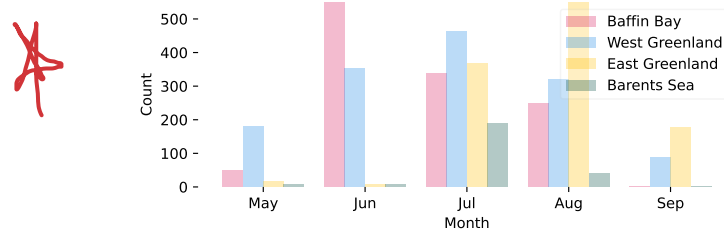


Figure 8: Detected and matched icebergs by month from 2016 to 2023, colored by region.

Sea (6%). The majority was sampled from June through August, and small subsets in May and September (Fig. 8). Most icebergs were observed at wind speeds below 10 m s^{-1} (Fig. 9a), and at positive air temperatures between 0° and 10° (Fig. 9b).

4.2. Iceberg area error

The iceberg areas were calculated from the Sentinel-2 images, and the apparent areas of detected icebergs were retrieved from the HH- and the HV-polarized Sentinel-1 images. Icebergs were detected either only in the HH channel (25% at the 10^{-6} PFA and 49-pixel OWS), only in the HV channel (20%), or in both channels (55%). To analyze differences in the area retrieval depending on the polarization channel, we divided the dataset into HH icebergs and HV icebergs. For the iceberg detection and area retrieval, we compared two window sizes and two PFAs (Tab. 4).

4.2.1. CFAR window size and PFA

A larger CFAR window size is beneficial for the area retrieval, as a small OWS restricts the area retrieval when the iceberg signature is larger than the window size (Fig. 10a,b). Fig. 10 shows results for the 10^{-6} PFA. If

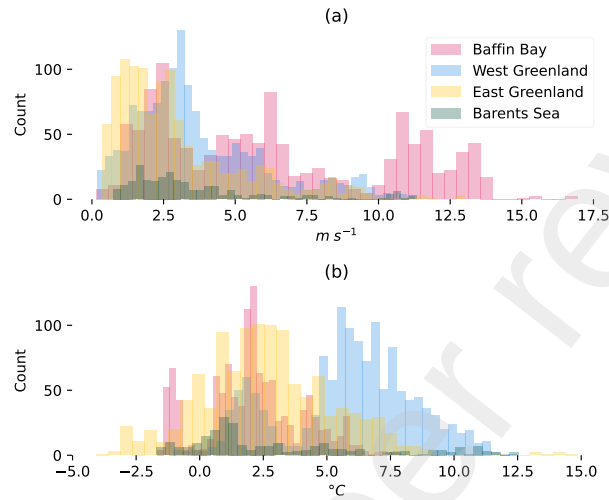


Figure 9: Wind speed (a) and air temperature (b) at which the iceberg pairs were obtained, colored by region.

Table 4: Number of icebergs sampled by polarization channel and PFA.

Category	PFA	Icebergs
HH icebergs	10^{-6}	3208
	10^{-12}	2879
HV icebergs	10^{-6}	2995
	10^{-12}	3020

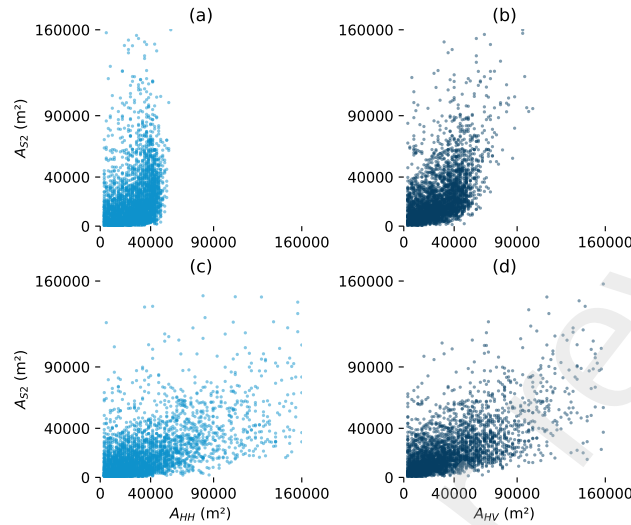


Figure 10: A_{HH} and A_{HV} plotted against A_{S2} for two OWS: 13 pixels (a, b), and 49 pixels (c, d).

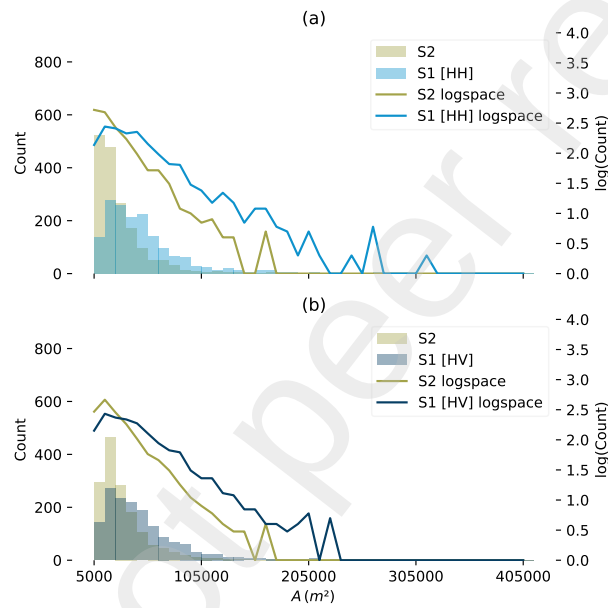
not specified otherwise, we examine the iceberg area retrieval for the 10^{-6} PFA with a 49-pixel OWS and a 35-pixel GWS. Independent of the OWS, iceberg areas are primarily overestimated, for small icebergs at a magnitude of several 100% (Tab. 5, Fig. 12).

4.2.2. Error analysis

The HH (HV) iceberg areas are equivalently correlated with the Sentinel-2 iceberg areas at a Pearson's r-value of 0.66 (0.69) (Tab. 5). The lower PFA reduces the MAE_A , while leaving the correlation with the Sentinel-2 iceberg areas mostly unaffected (Tab. 5).

The Sentinel-2 iceberg areas resemble a power-law distribution (Fig. 11a,b) with an average iceberg area of 24868 m^2 (RL: 140 m) and a maximum of 399700 m^2 (632 m). Given predominantly overestimated iceberg areas, the Sentinel-1 iceberg area distributions appear biased towards larger areas (Fig.

power law distr.
RL140 $\rightarrow A_A = 24868 \text{ m}^2$
632 $\rightarrow \text{max } 399700 \text{ m}^2$
(none < RL140?)



*S2
right skew*

Figure 11: Sentinel-1 iceberg area distributions for the HH channel (a), and the HV channel (b) with the corresponding Sentinel-2 iceberg area distributions. The line plots show the counts in log space, depicted on the right y-axes. The Sentinel-1 iceberg areas in this plot were derived at a 10^{-6} PFA and a 49-pixel OWS.

PFA	OWS	Channel	RE _A (% P10, P90)	MAE _A (m ²)	Pearson's r-value	Slope	Intercept (m ²)
10 ⁻⁶	13		-55, 500	18414	0.21	0.46	11082
10 ⁻¹²	13		-67, 390	17632	0.11	0.27	18484
10 ⁻⁶	49		-36, 569	27645	0.65	0.4	5487
10 ⁻¹²	49		-51, 454	21999	0.63	0.49	5996
10 ⁻⁶	13		-56, 250	15724	0.49	0.84	2422
10 ⁻¹²	13		-60, 220	14690	0.49	0.83	4911
10 ⁻⁶	49		-37, 349	23758	0.69	0.45	6556
10 ⁻¹²	49		-44, 302	20645	0.7	0.5	6604

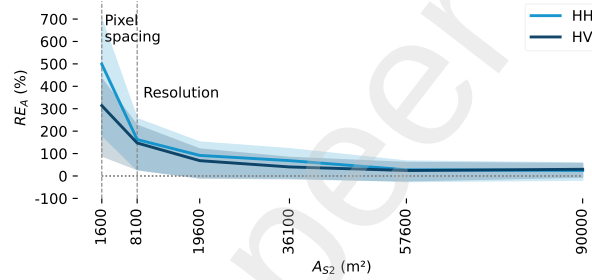


Figure 12: The mean RE_A in Sentinel-2 iceberg area bins for the 49-pixel OWS. The light-shaded areas depict the interquartile range of RE_A. The labels on the x-axis specify the lower limit in the Sentinel-2 area in each area bin. Pixel spacing and resolution refer to the Sentinel-1 EW GRDM data.

427 11).

428 Small icebergs exhibit large positive RE_A values with an increased spread
 429 in both polarization channels. The overestimation increases for icebergs
 430 smaller than the Sentinel-1 EW GRDM resolution (Fig. 12).

431 4.3. Error drivers

432 Subsequently, we analyze the effect of varying clutter, and iceberg backscat-
 433 ter on the SAR-based iceberg areas. Here, we aim to understand how such

434 variations relate to the relative error in the iceberg areas. We first quantify
435 relationships between backscatter statistics and the iceberg area error. We
436 show the relative error for *root lengths* RE_{RL} (Fig. 13) to simplify the rela-
437 tionships to linear correlations. The RE_{RL} is negatively correlated with the
438 iceberg backscatter and clutter at HH-polarization (Fig. 13a,c) at a Pearson's
439 r-value of -0.56 for the iceberg backscatter, and -0.64 for the clutter. The
440 varying iceberg backscatter is the primary driver of the HV contrast (Fig.
441 13a) due to nearly invariant HV clutter (Fig. 13b). As a consequence, un-
442 derestimated HV iceberg areas primarily occur at an HV iceberg backscatter
443 below -20 dB (Fig. 13d). Moreover, underestimated iceberg areas are primar-
444 ily seen at low HH and HV contrast (Fig. 13e,f), but the linear correlation
445 is weak. The error variation of the relative error with the incidence angle is
446 low (Fig. 13g,h).

447 For both channels, the relationships between backscatter variables and the
448 RE_{RL} are stratified by the Sentinel-2 iceberg area. A subset of small icebergs
449 shows high positive errors in both channels departing from the cluster.

450 The HH (HV) iceberg backscatter of detected icebergs is strongly (weakly)
451 correlated with the HH clutter, with a Pearson's r-value of 0.88 (0.26) (Fig.
452 14a). At a specific HH clutter level, the HH iceberg backscatter varies above
453 a clutter-driven minimum backscatter to the right of the sharp boundary
454 of the data cluster (Fig. 14a). Thereby, the iceberg backscatter introduces
455 contrast variation.

456 We additionally recognize a weak dependency between iceberg size and
457 backscatter in Fig. 14. Larger icebergs tend to exhibit a higher backscat-
458 ter than smaller icebergs. We will discuss this observation as an effect of

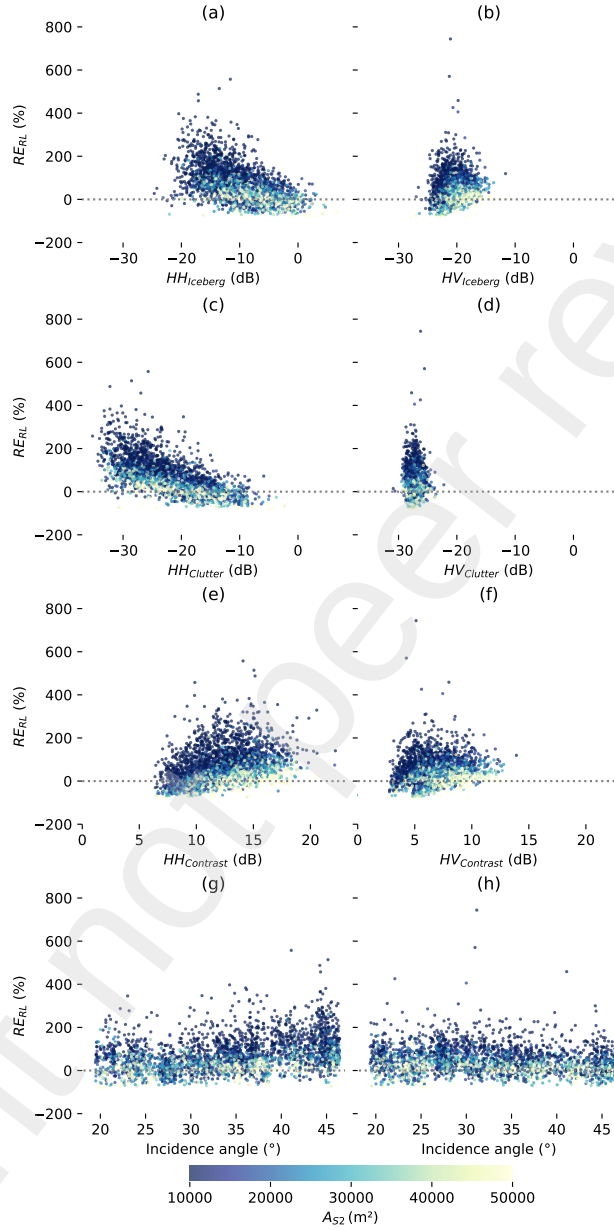


Figure 13: The iceberg backscatter (a, b), the clutter (c, d), the contrast (e, f), and the incidence angle (g, h) in the two channels (HH, HV) plotted against the RE_{RL}. To visualize the relation to size, the samples are colored by the Sentinel-2 iceberg areas.

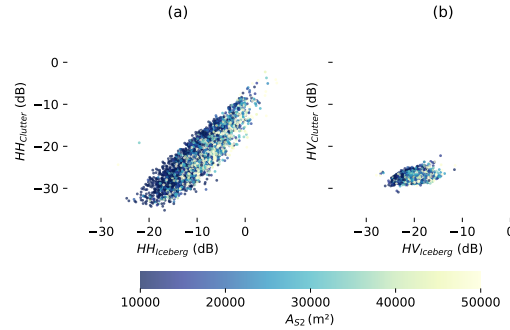


Figure 14: The iceberg backscatter versus clutter extracted in the outer CFAR window for HH icebergs (a) and HV icebergs (b). The samples are colored by the Sentinel-2 iceberg areas.

the iceberg size relative to the data resolution. Pearson’s r-values on the relationship between the iceberg backscatter and the Sentinel-2 iceberg *root lengths* are 0.34 (HH) and 0.26 (HV) (Fig. 15).

4.4. Iceberg area prediction

We evaluated the iceberg area models using the validation data unseen by the models during training. The aim was to find the best-performing model, referring to the combination of a feature set and a regression algorithm.

BackscatterRL models outperform *PureRL* models in all evaluation metrics. The best-performing models, *BackscatterRL* CB, reduce the MAE for the HH (HV) channel by 58% (49%) (Tab. 6). The *BackscatterRL* CB model yields a lower MAE and a stronger improvement compared to the CFAR-based iceberg areas for the HH channel than for the HV channel. This suggests that the iceberg area prediction in the HH channel benefits more from backscatter statistics and the incidence angle than in the HV channel.

Additionally, the CB models predict iceberg areas better than the classical

Table 6: Evaluation metrics of the iceberg area models. The best results are highlighted in bold. Metrics for the CFAR algorithm refer to the initial CFAR-based iceberg areas in the validation portion of the dataset. The other results show the metrics for predicted iceberg areas based on the validation portion.

	Feature set	Model	MAE _A (m ²)	RE _A (% , P10, P90)	Pearson's r-value
HH	CFAR		27778	-30, 537	0.69
	PureRL	OLS	15397	-44, 333	0.68
	PureRL	CB	15881	-55, 288	0.67
	BackscatterRL	OLS	12046	-49, 200	0.79
	BackscatterRL	CB	11767	-50, 166	0.78
HV	CFAR		24411	-35, 352	0.61
	PureRL	OLS	14771	-50, 215	0.61
	PureRL	CB	14904	-55, 189	0.58
	BackscatterRL	OLS	13317	-53, 177	0.66
	BackscatterRL	CB	12544	-50, 147	0.66

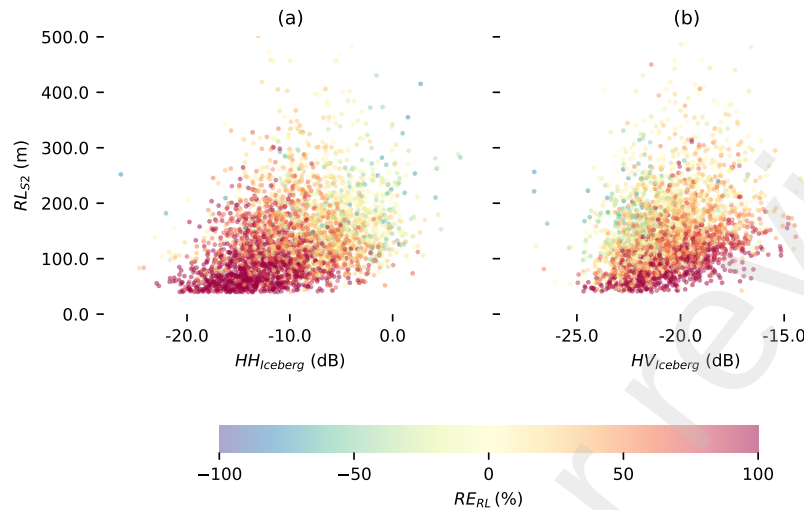


Figure 15: Mean iceberg backscatter by the Sentinel-2 *root length* for the HH channel (a) and the HV channel (b). To show the relation to the area error, the samples are colored by the RE_{RL} .

474 OLS. However, the backscatter-sensitive *BackscatterRL* feature set yields a
 475 stronger improvement than the regression algorithm choice at HH-polarization.
 476 That is, including backscatter information in the models is more beneficial
 477 than using a more advanced regression algorithm.

478 While all models yield improved iceberg areas across the iceberg size
 479 distribution, they also show a widening spread in the *Bias towards larger*
 480 *icebergs* (Fig. 16). Additionally, we recognize a slight tendency to *nega-*
 481 *tive biases for icebergs larger than 90000 m²*. This tendency is more pro-
 482 nounced for the OLS (Fig. 16e,f) than for the CB algorithm (Fig. 16g,h).
 483 The *BackscatterRL* CB models keep overestimating iceberg areas for icebergs
 484 smaller than 23700 m², whereas the relative error is around 0 for icebergs
 485 larger than that (Fig. 17b).

★ test

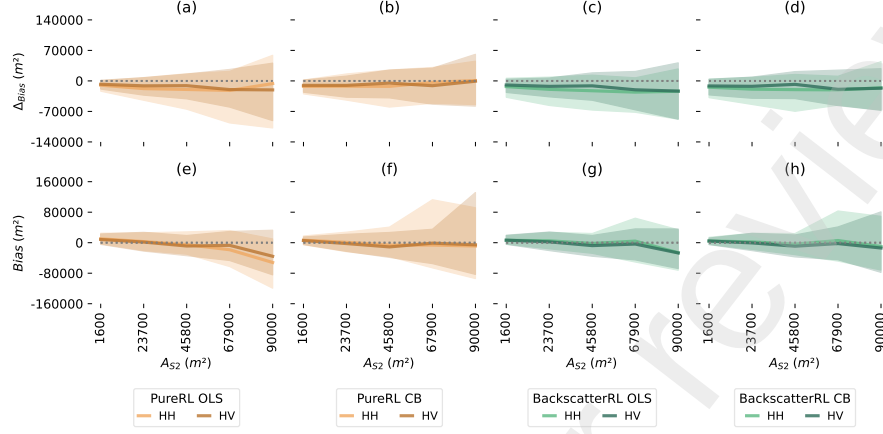


Figure 16: Δ_{Bias} and $Bias$ in A_{S2} area bins for the four models in two channels: *PureRL* OLS (a,e), *PureRL* CB (b,f), *BackscatterRL* OLS (c,g), *BackscatterRL* CB (d,h). Negative values of Δ_{Bias} indicate an improvement in the predicted iceberg areas compared to the initial CFAR-based iceberg areas. Negative (positive) values of the $Bias$ indicate underestimated (overestimated) iceberg areas. Values on the x-axis show the bottom limit of the area bin. The light-shaded areas depict the range between the 10th and the 90th percentile.

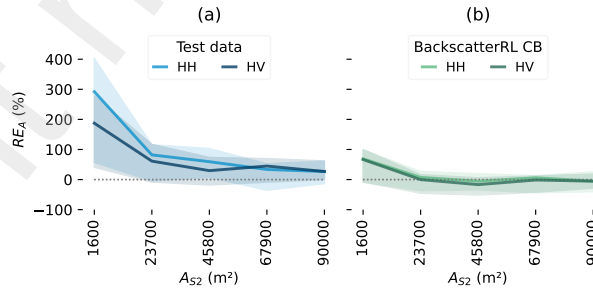


Figure 17: Mean RE_A for the test data (a), and the predictions by the *BackscatterRL* CB models. The light-shaded areas depict the interquartile range of RE_A .

5. Discussion

In this paper, we quantified errors in iceberg areas retrieved from Sentinel-1 SAR data using a CFAR algorithm, with reference to optical Sentinel-2 data. Further, we presented novel backscatter-sensitive regression and gradient boosting models predicting improved iceberg areas from Sentinel-1 data. The models use ocean clutter, iceberg backscatter statistics, and the SAR incidence angle in addition to the iceberg area delineated by a constant false alarm rate iceberg detector.

5.1. Drivers of the iceberg area error

We explain errors in CFAR-based iceberg areas by three effects (Fig. 18). First, we found a dependency between the iceberg size and its backscatter. We interpret this dependency partly as resolution-driven. While icebergs exhibit a naturally variable backscatter, we also need to take into account the spatial resolution of the SAR sensor relative to the iceberg size. Larger icebergs cover more pixels, whereas smaller icebergs cover fewer pixels or fractions of pixels, leading to a lower backscatter. Considering a range of iceberg sizes, we thus observe backscatter-size correlations. The natural and resolution-imposed iceberg backscatter variations affect the iceberg area retrieval because the CFAR algorithm separates a varying amount of pixels as outliers depending on the clutter (Fig. 18). Second, the ocean clutter varies with wind speed, especially in the HH channel, amplifying the variation in the retrieved apparent iceberg areas. Third, the CFAR clutter statistic can be contaminated by ice features in both polarization channels, adding variation to the CFAR-based threshold (Fig. 18). CFAR algorithms were developed

510 to adapt to these variations. The flexibility affects the iceberg area retrieval
511 because the iceberg area shrink as the clutter (iceberg backscatter) increases
512 (decreases). In the extreme case, the clutter (iceberg backscatter) is too high
513 (too low) to detect the iceberg, resulting in zero area. It is important to
514 understand that, in the HH channel, the clutter and the iceberg backscatter
515 of CFAR-detected icebergs are connected because the clutter and the iceberg
516 both drive the contrast, and a minimum contrast has to be fulfilled for flag-
517 ging an outlier. The area retrieval in the HV channel depends more on the
518 iceberg backscatter than on the clutter, since the clutter at HV-polarization
519 is very low and almost independent of wind speed.

520 **Clutter contamination** is a well-known limitation of CFAR algorithms
521 (Tao et al., 2016). *self-contamination* affects the iceberg area retrieval ad-
522 versely when the CFAR window is too small relative to the present icebergs
523 (Fig. 18). Enlarged iceberg signatures in the HH channel (Fig. 19a,e), which
524 we observed across our dataset, further complicate the area retrieval. The
525 surface backscattering intensity is significantly stronger at HH- than at HV
526 polarization, which could also explain why, in turn, small icebergs were de-
527 tected more frequently in the HH channel. Volume scattering is reduced or
528 even suppressed during melting conditions at both polarizations (Færch et
529 al., 2024). The enlarged iceberg signatures in the HH channel amplify the
530 self-contamination effect when the OWS is too small.

531 The outer window should be sufficiently large, but we expect the clutter
532 to decorrelate with the direct surrounding of a specific iceberg as the window
533 size increases. Therefore, we proposed the 49-pixel OWS with a 35-pixel
534 GWS ($1400 \cdot 1400$ m) for the iceberg sizes covered in our study. The 49-

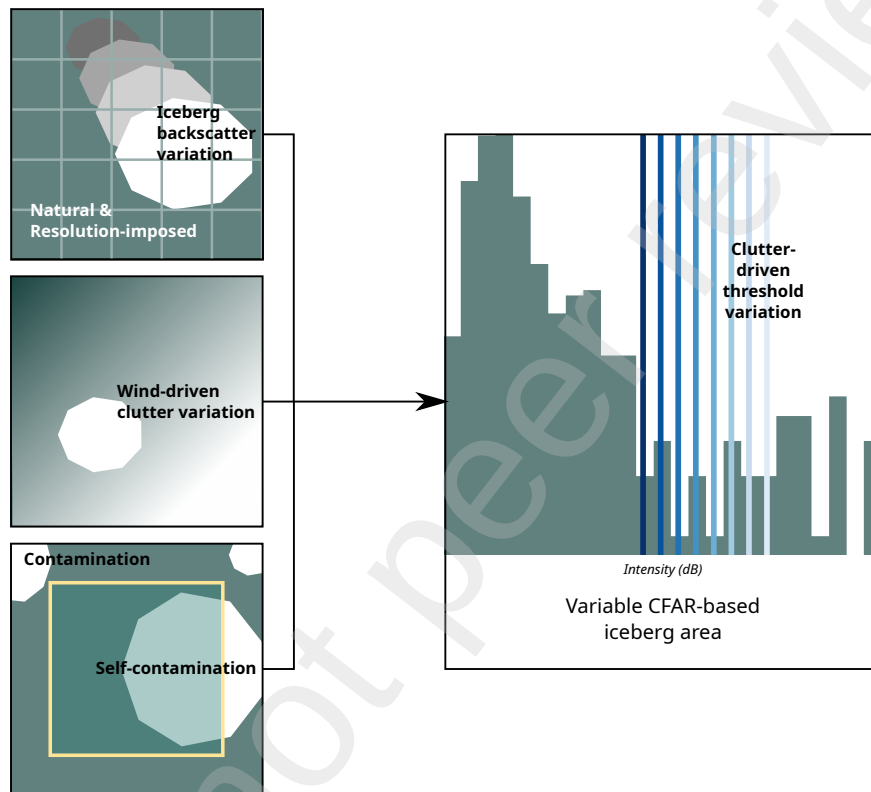


Figure 18: Three effects explaining the errors in CFAR-based iceberg areas. The iceberg backscatter varies naturally and as a function of the SAR spatial resolution. The clutter varies with wind speed, particularly in the HH channel. Large icebergs and neighboring icebergs can contaminate the clutter statistics. In combination, these variations cause a variable CFAR-based iceberg area and related errors. We utilized these effects indirectly in our area prediction models to yield improved iceberg areas.

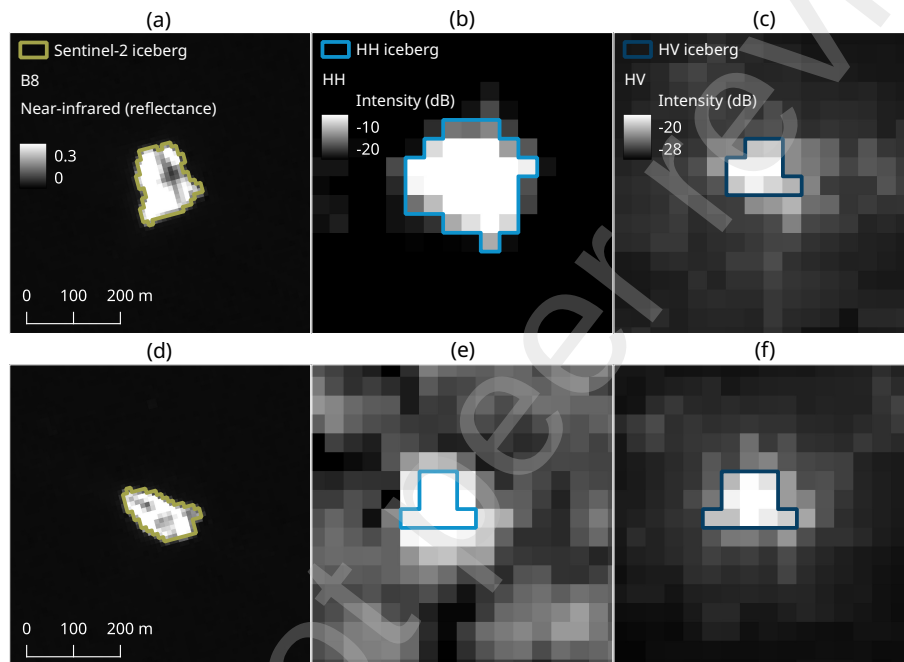


Figure 19: These examples visualize the iceberg backscatter variation and the wind-driven clutter variation. The two icebergs are shown in the Sentinel-2 near-infrared band (a, d), and in the two Sentinel-1 channels (b, c, e, f) on 18 June 2021 in the Davis Strait off the coast of West Greenland. The top iceberg was detected at $76.750289^{\circ}\text{N}$, $76.050145^{\circ}\text{W}$ at an incidence angle of 24° , the bottom at $76.5990367^{\circ}\text{N}$, $75.9788113^{\circ}\text{W}$ at an incidence angle of 23° .

535 pixel OWS reduces self-contamination, but wind-driven clutter variations,
536 and variations in the iceberg backscatter still affect the CFAR-based iceberg
537 areas.

538 Statistics of the clutter, iceberg backscatter, and iceberg contrast improve
539 the iceberg area prediction, using our *BackscatterRL* CB models. To under-
540 stand better how the models predict iceberg areas, we obtained the feature
541 importance metric of the CB models. The feature importance for the CFAR
542 iceberg area is roughly 60% in both channels. The other 40% of the pre-
543 diction relies on backscatter statistics and the incidence angle. The feature
544 importance for the HH model indicates an 9% contribution of the HH clutter,
545 whereas it is 4% for the HV model. This suggests that the HH model utilizes
546 the clutter-dependency of the CFAR-based area retrieval. As expected, the
547 HH iceberg areas benefit more from the backscatter-sensitive area prediction
548 than the HV iceberg areas. While the CFAR-based HH iceberg areas are
549 initially more erroneous, there is a systematic connection between the HH
550 clutter and the error.

551 Overall, the improvement associated with the backscatter-sensitive mod-
552 els is substantial. However, the spread in the bias increases with iceberg
553 size. This could be due to the fewer unique samples available for very large
554 icebergs. Over- and undersampling balanced the dataset, but the more abun-
555 dant iceberg size bins still contain more information. The OLS tends to un-
556 derestimate iceberg areas for icebergs larger than about 300 m RL, whereas
557 this tendency is less pronounced in the gradient boosting models. This may
558 be a sign that the OLS does not depict non-linear relationships present in
559 the data, whereas gradient boosting is able to learn those relationships.

560 5.2. Limitations of the study

561 The iceberg area retrieval from SAR data had received sparse attention
562 prior to our study. Although Buus-Hinkler et al. (2016) quantified errors
563 in the Sentinel-1 iceberg area retrieval with reference to Landsat data, the
564 dataset was probably too small to reliably systematically explain the errors.
565 We attempted to explain and mitigate errors in Sentinel-1 iceberg areas, using
566 thousands of Optical-SAR iceberg pairs observed under various wind speeds
567 across the Arctic. Our dataset was limited to open water, mostly positive air
568 temperatures, and wind speeds below $15 \text{ m} \cdot \text{s}^{-1}$. The required Optical-SAR
569 iceberg matching possibly introduced a limit in represented wind speeds,
570 since longer drift patterns impede the matching. We consciously limited
571 the study to open water due to the difficulties in distinguishing icebergs
572 from deformed sea ice in C-band SAR in the summer months (Færch et al.,
573 2024). Furthermore, while matching Optical-SAR iceberg pairs in land-fast
574 ice is feasible, it becomes very challenging in high drift ice concentrations.
575 Generally, high ice concentrations should affect the iceberg area retrieval from
576 HH data similarly as wind-roughened water. However, the HV backscatter
577 of sea ice is higher than for open water, hence, we expect a larger difference
578 to our results presented here.

579 Our study covered diverse conditions by using co-located optical satellite
580 data as a reference. Although optical satellite data at higher resolutions exist,
581 and airborne or drone data would provide even higher resolutions, the less
582 systematic acquisition of these data impedes the matching with SAR. While
583 we should keep in mind the inherent errors in remotely retrieved iceberg areas,
584 we argue that Sentinel-2 data provide a viable trade-off between resolution

limited to
open
water
 $T > 0^\circ\text{C}$
 $WS < 15 \text{ m/s}$

diverse
conditions
dataset?

585 and temporal coverage.

586 5.3. Implications for SAR-based iceberg areas

587 In this study, we developed a backscatter-sensitive gradient boosting
588 method for retrieving iceberg areas from Sentinel-1 EW data. We demon-
589 strated that counting the number of bright pixels belonging to an iceberg
590 often provides an inaccurate idea of the iceberg area, even when done visu-
591 ally (Fig. 19). This raises questions regarding a suitable training of more
592 advanced object detection algorithms, such as convolutional neural networks,
593 for the SAR-based iceberg detection and area retrieval. Our results indicate
594 that training such algorithms on the apparent iceberg boundaries (Fig. 19)
595 is likely to yield inaccurate iceberg areas. In the light of ambiguous iceberg
596 boundaries, we suggest treating iceberg detection and the area retrieval as
597 separate problems that might require individual algorithmic solutions. Gen-
598 erally, a higher data resolution may reduce resolution effects, but the variable
599 scattering of icebergs remains a challenge. In this paper, we compared indi-
600 vidually matched icebergs, but in future work, we propose to also evaluate
601 Sentinel-1-based area distributions and total iceberg areas. Furthermore, we
602 suggest studying the iceberg area retrieval from SAR data at different fre-
603 quencies and resolutions. Possible future work may include examining the
604 iceberg area retrieval for icebergs in sea ice, possibly covering winter months
605 with C- and L-band data, and summer months with L-band data only (Færch
606 et al., 2024).

607 5.4. *Implications for applications on iceberg areas*

608 Accurate iceberg areas are relevant to applications related to maritime
609 safety, oceanography, and glaciology. Applications on statistical iceberg risk
610 base their modeling on observational iceberg statistics, e.g., derived from
611 satellite data. These studies typically require above-waterline iceberg area
612 distributions to estimate the iceberg keel depth (Monteban et al., 2020).
613 Geophysical applications need iceberg area distributions, and total above-
614 waterline iceberg areas integrated over the distributions (Enderlin et al.,
615 2016; Sulak et al., 2017; Moyer et al., 2019; Scheick et al., 2019; Rezvan-
616 behbahani et al., 2020; Shiggins et al., 2023). In our view, the compli-
617 cated interpretation of SAR data has limited the take-up of these data for
618 quantifying iceberg areas in geophysical applications. With the presented
619 backscatter-sensitive area retrieval from Sentinel-1 EW GRDM data, we aim
620 to provide a tool for retrieving iceberg areas from SAR data in open water
621 during summer months. Further model validation should also take into ac-
622 count varying ice concentrations and sea ice presence, particularly in glacial
623 fjords.

624 6. Conclusions

625 We presented a new method for predicting above-waterline iceberg areas
626 from Sentinel-1 EW GRDM data in the HH and HV polarization channel.
627 Our study was based on a dataset of 4014 icebergs in open water detected and
628 matched in co-located Sentinel-1 SAR images and optical Sentinel-2 images
629 across the Arctic. To conclude:

- 630 1. Gradient boosting regression models predict iceberg areas more accu-
631 rately when trained with ocean clutter, the iceberg backscatter, the ice-
632 berg contrast, and the SAR incidence angle, in addition to the CFAR-
633 derived iceberg area. The *BackscatterRL* gradient boosting models
634 outperform the other models.
- 635 2. The relative error in CFAR-derived iceberg areas retrieved from the
636 HH channel is negatively related to HH ocean clutter. Independent of
637 the channel, underestimated iceberg areas predominantly occur when
638 the contrast between the iceberg and its surrounding is low.
- 639 3. The iceberg backscatter is positively related to the iceberg area, which
640 we explain primarily as an effect of the spatial resolution.
- 641 4. Our results apply to icebergs with a maximum area of about 400000 m²
642 in open water at positive air temperatures in the months May through
643 September.

644 7. Author contributions

645 Henrik Fisser: Conceptualization, Methodology, Software, Validation,
646 Formal analysis, Investigation, Resources, Data Curation, Writing - Original
647 Draft, Writing - Review & Editing, Visualization. Anthony P. Doulgeris:
648 Methodology, Writing - Review & Editing, Supervision, Project adminis-
649 tration, Funding acquisition. Wolfgang Dierking: Methodology, Writing -
650 Review & Editing.

651 8. Data and code availability

652 The *BackscatterRL* CB model is available as a Python package on GitHub:
653 [S1.IcebergArea](#). The dataset compiled and analyzed in this study is avail-
654 able on Mendeley Data: Fisser, Henrik (2024), “Sentinel-1 Iceberg Areas”,
655 Mendeley Data, V1, doi: 10.17632/z9jdmxrkv.1.

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